Annie

Montgomery McLaughlin Gallot Woltman algorithm

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Notation:
"mod M" is the operator: $y = x \mod M \Rightarrow y \in [0, M-1]$.
The congruence relation is noted $a \equiv b \pmod{M}$.

Montgomery modular multiplication

Peter L. Montgomery presented a method for multiplying two integers modulo N while avoiding division by N [3].

Theorem 1. *Let N be a positive integer, $R > N$, $\gcd(R, N) = 1$, $0 \leq x, y < N$. If*

(1) $T = x\,y$,

(2) $m = -T\,N^{-1} \mod R$,

(3) $r = (T + m\,N)\,/\,R$,

(4) $t = r - N$ *if* $r \geq N$ *and* $t = r$ *otherwise*

then $t = R^{-1}\,x\,y \mod N$.

Proof. (2) $\Rightarrow m\,N \equiv -T \pmod{R}$ then r is an integer. (3) $\Rightarrow r \equiv T\,R^{-1} \pmod{N}$.
$0 \leq m\,N < R\,N$ and $0 \leq T < N^2 < R\,N$ then $0 \leq r < 2\,N$. □

In practice, x is converted into Montgomery form $x^* = R\,x \mod N$ (y, z resp.).
We have $z = x\,y \mod N \Leftrightarrow z^* = R^{-1}\,x^*\,y^* \mod N$. Conversion out of Montgomery form is done by computing $x = R^{-1}\,x^* \mod N$.

Montgomery modular multiplication without test

The test (4) can be removed if $R > 4\,N$ and $0 \leq x, y, r < 2\,N$.

Theorem 2. *Let N be a positive integer, $R > 4\,N$, $\gcd(R, N) = 1$, $0 \leq x, y < 2\,N$. If*

(1) $T = x\,y$,

(2) $m = -T\,N^{-1} \mod R$,

(3) $r = (T + m\,N)\,/\,R$

then $r \equiv R^{-1}\,x\,y \pmod{N}$ *and* $0 \leq r < 2\,N$.

Proof. $0 \leq m\,N < R\,N$ and $0 \leq T < 4\,N^2 < R\,N$ then $0 \leq r < 2\,N$. □

McLaughlin modular multiplication (without test)

Philip B. McLaughlin Jr. extended the method using a second integer Q [2].

Theorem 3. *Let N be a positive integer, $R > 4N$, $Q > R$, $\gcd(R, N) = \gcd(Q, R) = 1$, $0 \le x, y < 2N$. If*

(1) $T_R = x\,y \mod R$, $T_Q = x\,y \mod Q$,

(2) $m = -T_R N^{-1} \mod R$,

(3) $r = (T_Q + mN)R^{-1} \mod Q$

then $r \equiv R^{-1}\,x\,y \pmod N$ *and* $0 \le r < 2N$.

Proof. $Q > 2N$ and r in $[0, Q-1]$ is unique. Then r is the solution. □

Gallot & Woltman method

The following method has emerged from a discussion on the mersenneforum [1].

$Q = R + 2$ is fixed. R must be odd such that $\gcd(Q, R) = 1$.
Because $R^{-1} = -2^{-1} \mod Q$, we define the new form

$$x^* = 2^{-1} R\,x \mod N$$

(y, z resp.). Now, we have $z = x\,y \mod N \Leftrightarrow z^* = 2R^{-1}\,x^*\,y^* \mod N$.
Note that N must be odd such that $2^{-1} \mod N$ exists. If $T_R \neq 0$ and $m' = T_R N^{-1} \mod R$ then $m = R - m' = -(m' + 2) \mod Q$. If $T_R = 0$ then $T_Q = m' = 0$ and $r = N$.
Finally, if $0 \le x < 2N$ then $0 \le r \le 4N$. New conditions are needed:

Theorem 4. *Let N and R be positive odd integers, $\gcd(R, N) = 1$, $R > 16N$, $Q = R + 2$, $0 \le x, y < 4N$. If*

(1) $T_R = x\,y \mod R$, $T_Q = x\,y \mod Q$,

(2) $m' = T_R N^{-1} \mod R$,

(3) $r' = \left((m' + 2)N - T_Q\right) \mod Q$

then $r' \equiv 2R^{-1}\,x\,y \pmod N$ *and* $0 \le r' < 4N$.

Proof. We have $r' = 2r = 2(T + mN)/R$. $0 \le mN < RN$ and $0 \le T < 16N^2 < RN$ then $0 \le r' < 4N$. □

The implementation in Annie uses $R = 2^n - 1$. The polynomial multiplication modulo R is a cyclic (positive-wrapped) convolution and modulo Q is a negacyclic (negative-wrapped) convolution.
Number Theoretic Transforms achieve $O(n \log n)$ time complexity (they are a divide and conquer technique).
The code for the cyclic and negacyclic convolutions is shared. The twiddle factors are the nth root of unity for R and the nth root of -1 for Q.

References

[1] Yves Gallot, George Woltman *et al.*, *A new algorithm for generic modular reduction*, mersenneforum.org, Aug.–Sep. 2023, https://www.mersenneforum.org/node/22013.

[2] Philip B. McLaughlin Jr., *New frameworks for Montgomery's modular multiplication method*, Math. Comp. **73** (2004), 899–906, DOI: https://doi.org/10.1090/S0025-5718-03-01543-6.

[3] Peter L. Montgomery, *Modular multiplication without trial division*, Math. Comp. **44** (1985), 519–521, DOI: https://doi.org/10.1090/S0025-5718-1985-0777282-X.